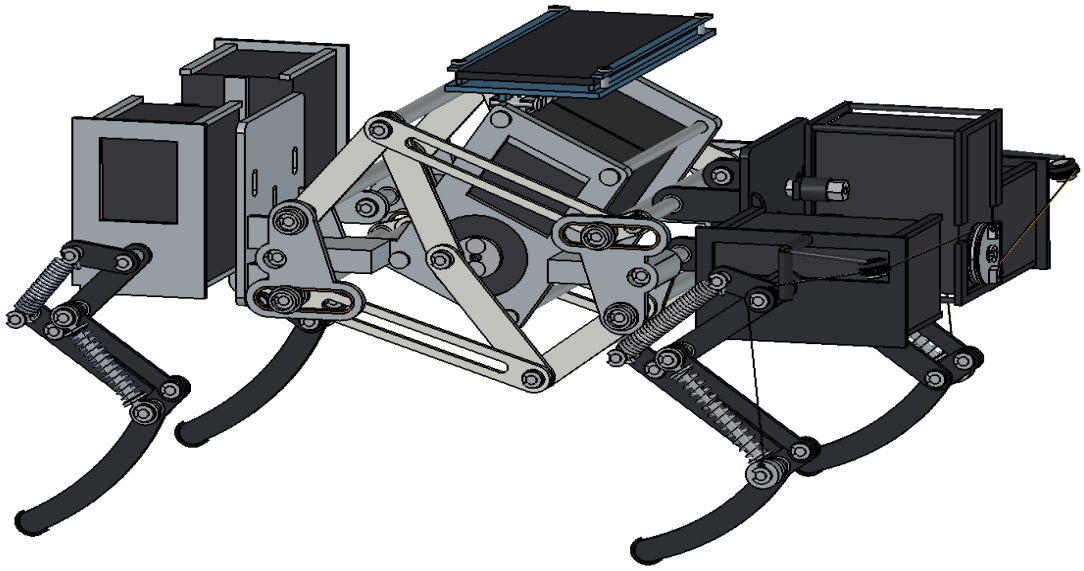
Assembly & calibration record

Review candidate r07 | FLEXOR-F inspired active-spine quadruped

REVIEW CANDIDATE	PROVISIONAL INTERFACES	NO HARDWARE OPERATION AUTHORIZED



Actual saved FreeCAD view of the r07 candidate. Motor and OpenRB blocks are envelopes; this image does not establish supplier mount or output datums.

486 physical links	105 BOM part numbers	268.376 g known nominal mass	1 unknown-mass board

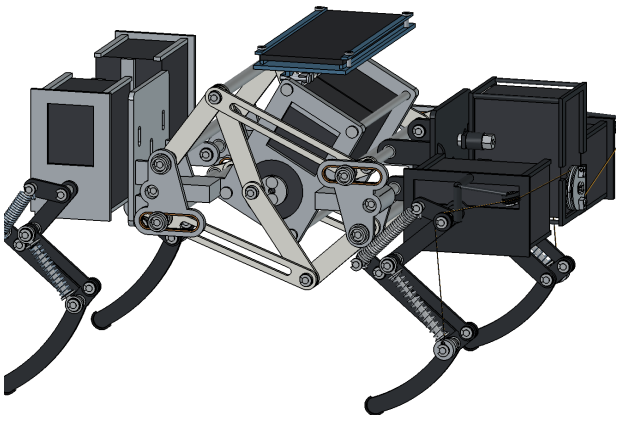
This document describes the saved candidate, a practical assembly order, a proposed six-coordinate motor map, a torque-off calibration record, nominal demand screens and every BOM part number. It does not authorize fabrication, energizing, device-ID writes, motion tests or load tests.

1 | Architecture and source fidelity

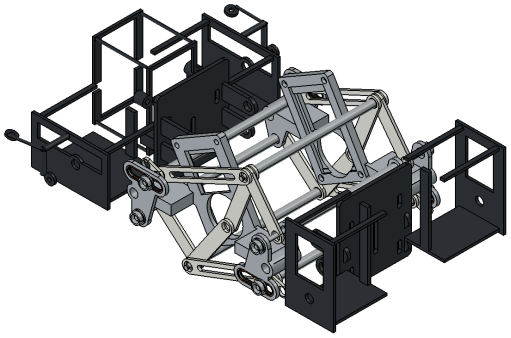
The modeled FLEXOR-F inspired candidate uses two XL330-M077-T motors redundantly driving one spine phase, four XC330-M181-T proximal leg motors and one XC330-M181-T hind-knee wheel motor. Six coordinates are commanded: one shared spine phase, four hip/thigh coordinates and one hind tendon-wheel coordinate. Foreleg elbow behavior and hind knees remain passive in the model. The assembly keeps the coupled two-output spine, three-segment limbs, paired hind cable route, and separate extension/compression spring representations.

Source basis and interpretation

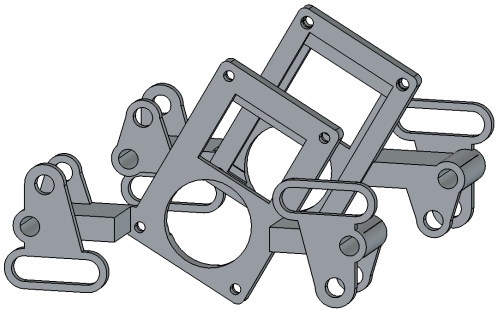
The main paper (pp. 4 and 16) and supplement (Table S1, p. 65) establish the broad actuator count and mechanism topology; Movie S1 (51-80 s) establishes visible arrangement and morphology, not precision dimensions. The r06-to-r07 frame interpretation follows the animation: a rounded lower lumbar opening, an upper window, local end shelves, short bearing ears, and distinct fore/hind transverse torso plates. Numeric cradle, plate, hole, fit, bracket and support dimensions are reconstruction assumptions unless separately supplier-verified. The current r07 supplier review confirms that exact motor mounting/output/horn and populated OpenRB interfaces remain unresolved. Source registry and frame interpretation: docs/01_reference_evidence.md and docs/13_candidate_frames_r06.md.



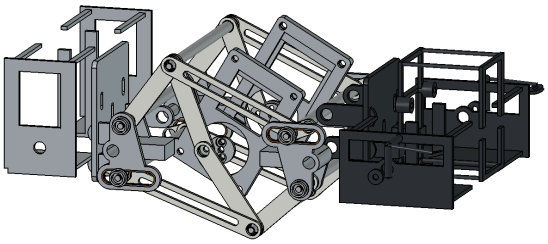
Assembled side view



Spine and torso frames



Lumbar cradle/window



Source-facing spine detail

Screenshots are actual FreeCAD captures of saved r07 parts and placements. They are not photographs of a built robot.

Scale and envelope comparison

Native B-rep bounds are 164.3376 x 119.6 x 104.0254 mm versus the 110 x 99 x 84 mm reference (+54.3376, +20.6000, +20.0254 mm). Preserve the mismatch for design review; do not scale the assembly to force a match.

2 | Assembly sequence

Follow the saved native part placements and the exact PN list in Appendix A. Record measured purchase-part identity and dimensions before finalizing any mating part.

1. Stage and inspect parts

Match items by exact BOM part number. Record vendor code, batch, material, coating, measured dimensions and mass. Keep the controller mass unknown until verified.

2. Build the spine core

Assemble A/B/C shaft families, native links, cranks, copper guides/liners and retainers in the saved arrangement. Preserve both coupled outputs and the two-motor spine drive. Check axial capture, free motion and tool access by inspection.

3. Fit lumbar and torso frames

Install left/right rounded lumbar cradles and fore/hind transverse torso frames. Retain the saved shaft and tie topology. Do not substitute a flat center plate or add hardware at unmodeled positions. Inspect the upper window, lower opening, local shelves and guide ears.

4. Install spine motors and adapters

Locate the XL case envelopes as modeled. Mate horns/cranks only after matching the exact motor output drawing or measuring a sample. Verify output center, spline, retaining screw, case holes and screw engagement independently.

5. Assemble four legs

Keep all leg frames in the model convention: +X forward, +Y left, +Z up. Fit hip axes, three segment bodies, pins, bearings, spacers, clips and return features in the saved side. Fore/hind lengths differ; left/right retain the same sagittal angle convention.

2 | Assembly sequence (continued)

6. Fit springs and hind tendon drive

Install one extension and one compression spring per leg with actual free length, rate, end form and preload recorded. Keep hind knees passive in the model. Seat each cable at its positive-capture feature; validate cable construction, clamp pressure, bend radius, tension retention and end slip on separate coupons before loading.

7. Fit controller carrier

Place the OpenRB envelope in the free-edge tray, add the edge retainers and close both split clamps around the existing nominal 2.4 mm fixed ties. The 2.5 mm bore gives 0.10 mm nominal diametral clearance; clamp preload, friction, creep and retention are not qualified. Do not drill or assume PCB holes.

8. Route harnesses and inspect service access

Route a separate TTL daisy-chain and external power harness with chassis strain relief and service loops clear of shafts, springs, tendons, bearings and clips. Use actual connector/cable dimensions. Check plug access and removable retainers.

9. No-power mechanical inspection

With torque off and external supply disconnected, inspect fastener engagement, clip seating, free joint movement where possible, spring seating, cable paths, pinch points, polarity labels and connector clearance. Stop at binding or mismatch. This is a proposed inspection order; physical hardware inspection has not occurred.

Nominal stackups to check against real parts

Interface	Nominal stack / result	Unresolved
Bearing seat	4.05 mm seat / 4.00 mm bearing OD = 0.05 mm diametral clearance	Vendor bearing type, tolerance and process size
Shaft/bearing bore	1.90 mm shaft / 2.00 mm bore = 0.10 mm diametral clearance	Shaft finish, coaxiality and running fit
Controller tie clamp	2.50 mm clamp bore / 2.40 mm tie OD = 0.10 mm clearance	Preload, seating friction, creep and retention
PCB edge tray	Board 25 x 66 x 1.6 mm; tray 28 x 70 mm; 1.5 mm nominal each side	Board thickness and component-free edges assumed; population volume reserved separately

3 | Controller map and torque-off calibration record

The following is a proposed logical ID map only. No IDs were written, no servo offsets measured, and no powered test or hardware movement was performed.

Candidate ID	Device	Commanded coordinate
1	XL330-M077-T spine motor A	Shared spine phase theta
2	XL330-M077-T spine motor B	Same theta; redundant pair
3	XC330-M181-T fore left	Fore-left hip/thigh
4	XC330-M181-T fore right	Fore-right hip/thigh
5	XC330-M181-T hind left	Hind-left hip/thigh
6	XC330-M181-T hind right	Hind-right hip/thigh
7	XC330-M181-T hind tendon wheel	Bilateral tendon take-up

Six commands result because IDs 1 and 2 share one spine coordinate. Foreleg passive motion and both hind knees remain passive. The left/right legs use the same modeled positive sagittal direction; record any real motor sign inversion per ID instead of changing the model convention.

Power interface hold point

Candidate bench architecture: regulated external 5 V rail, TTL half-duplex bus and common ground. Confirm the OpenRB input, motor-rail separation, current capacity, connector pinout and USB back-feed from official documentation before connecting. USB is not a motor power supply. This document does not authorize energizing, device-ID writes, torque enable, jogging, a gait run, cable pull or calibration motion.

Record-only calibration checklist (torque off)

ID / pair	Record before any later authorized commissioning
Each ID 1-7	Exact model and serial; measured output-to-link orientation; direction sign; constrained reference; raw encoder reading; selected offset; software limits; current/temperature limits; operator and date.
Spine pair 1/2	Record both mechanical references and unloaded tracking error. Establish safe tracking, torque-sharing and stop limits before enable.
Leg IDs 3-6	Verify the same positive sagittal direction at all four mounts and record each installation sign.
Wheel ID 7	Verify winding sign independently for both cable branches and record terminal retention evidence.

Saved neutral pose is a software reference only; it does not establish encoder zero, spline tooth orientation, mechanical stop, tendon preload or loaded spring equilibrium. Leave offsets/signs unset and torque disabled at startup until actual parts and safe constraints are identified.

4 | Mass, center of mass and inertia estimate

Read-only FreeCAD calculation from the saved 486-link assembly; each link mass comes from nominal PurchasedMass_g or B-rep volume times the r07 material density estimate. The point-mass inertia omits each solid's own centroidal inertia.

Quantity	Known mass case	Separate sensitivity case
Mass	268.3758 g	286.3758 g
Center of mass X/Y/Z	(-8.6965, -0.0006, 3.1116) mm	(-8.7784, -0.0006, 5.4554) mm
Point-mass inertia diagonal Ixx/Iyy/Izz	220317.59 / 634978.84 / 777068.56 g mm^2	243772.09 / 658462.01 / 777097.22 g mm^2

Point-mass inertia tensor rows/columns X, Y, Z (g mm^2)

Case	Tensor
Known mass	[[220317.59, -25.51, -19292.68], [-25.51, 634978.84, -4.77], [-19292.68, -4.77, 777068.56]]
Separate sensitivity	[[243772.09, -25.49, -18472.76], [-25.49, 658462.01, -5.17], [-18472.76, -5.17, 777097.22]]

The known sum contains 485 mass-bearing links. The OpenRB PCB envelope is the sole unknown-mass item and stays unknown in the BOM. The sensitivity adds 10 g controller + 8 g wiring from candidate_r02.json, both lumped at actual board-envelope center (-10.000, -0.000, 40.400) mm. These assumptions do not change the BOM. Neither mass target nor reference envelope is demonstrated; no physical weighing was done.

Quasi-static demand screen

Loads use the separate 286.3758 g sensitivity case, gravity, alpha = 1/3/6, two supports with equal half-weight shares or one-leg full-weight worst case, and horizontal force Fx = 0.30 Fz. Neutral-pose lever arms use actual hip-journal centers and an estimated foot point from the lowest vertices of each native foot solid.

Alpha	Support	Fore hip N m	Hind hip N m	Pin shear MPa	Pin bend MPa	Seat pressure MPa	B shaft MPa
1	2-leg equal	0.0323	0.0378	0.52	8.55	0.31	4.60
1	1-leg worst	0.0646	0.0756	1.03	17.10	0.61	9.20
3	2-leg equal	0.0969	0.1134	1.55	25.64	0.92	13.80
3	1-leg worst	0.1939	0.2269	3.10	51.29	1.83	27.60
6	2-leg equal	0.1939	0.2269	3.10	51.29	1.83	27.60
6	1-leg worst	0.3878	0.4537	6.21	102.58	3.67	55.20

Demands only, not material allowables or PASS results. Pin stress uses d=1.9 mm, single-shear area, and a 3.925 mm native bearing-to-joint-plane overhang; bending is a simplified circular beam estimate. Seat pressure uses 4.0 x 1.2 mm projected bearing area. Spine B shaft uses a simply-supported central point load, dia 3.6 mm and assumed 60 mm maximum span. Contact distribution, support reaction sharing, dynamics and local failure modes are not solved.

At alpha 6, one-leg hip moments are 65% fore and 76% hind of the published XC330-M181-T 0.60 N m stall torque at 5 V. Stall is not a continuous-duty limit; no thermal, speed, duty-cycle, impact or fatigue margin is shown. Cable terminal force/friction and all structural allowables remain unknown.

5 | Interfaces, open items and release status

Supplier dimensions and physical fits

ROBOTIS text confirms the XL/XC nominal motor case envelope and lists drawings, but exact output center, case mounting pattern, horn mating geometry and connector-inclusive bounds were not obtained. OpenRB published board outline is 25 x 66 mm; thickness, mounting holes, populated component dimensions and connectors remain unverified. Do not treat placeholder motor or board solids as complete vendor CAD.

Nominal candidate stackups requiring metrology: 4.05 mm bearing seat / 4.00 mm OD gives 0.05 mm diametral clearance; 1.90 mm shaft / 2.00 mm bearing bore gives 0.10 mm; controller clamp bore 2.50 mm / fixed tie OD 2.40 mm gives 0.10 mm. The PA12 bearing boss nominal outer dia. 7.00 mm gives a 1.475 mm radial wall. These are geometry assumptions, not tolerance limits.

Spring parameters are assumptions: extension free length 14 mm and rate 0.25 N/mm; compression free length 18 mm and rate 0.4 N/mm; compression solid length 6 mm plus 1 mm bind margin. Catalogue identity, preload, fatigue, end form and measured lengths are open. Positive cable-capture geometry is modeled; required grip force, surface preparation, clamp torque, friction and slip limit are unverified.

Verification boundary

The current primary-MCP check sampled the C1/C2 moving frame pair against two frame pieces at 361 states and reported PASS with a 0.35 mm minimum gap. This is a scoped pairwise clearance result only. Full whole-robot motion remains unqualified; the check does not establish whole-assembly G3, loaded equilibrium, strength or manufacturing accuracy. The earlier 472-link frame/clamp report is a partial integration only. This manual makes no G3, G4 or G5 pass claim. Fabrication release is not established.

Open release blocker
- Revision-matched motor mounting/output/horn drawings and populated connector clearance.
- Measured OpenRB board thickness, component/connector keepouts, mounting strategy and mass.
- Bearing, shaft, clip, screw engagement and printed-fit tolerance stackups.
- Spring catalogue/rate/free-length/preload and cable terminal holding/slip evidence.
- Material/process allowables, safety factors, frame load-path review and full assembly collision/access evidence.
- Final drawing/export checks and physical build inspection.

Evidence files

Native candidate: cad/flexor_r07_robot.FCStd. BOM: bom/bom_r07.csv and bom/instances_r07.csv. Mass and envelope: reports/verification/r07_mass_envelope.json. Mass/CoM/inertia/load source: reports/verification/r07_engineering_screen.json. Current scoped 361-state C1/C2-vs-two-frame PASS is reported by the primary MCP review; whole-robot motion is unqualified. Earlier partial report: reports/verification/r07_frame_clamps_neutral.json. Supplier evidence: reports/sources/supplier_interfaces.md and reports/sources/supplier_completion_r07.md. Actual screenshots used here: reports/views/r07_robot_assembled.png, r07_robot_side.png, r07_spine_torso_frames.png, r07_bare_lumbar.png, r07_spine_source_facing.png.

Primary context: Wang et al., Science Advances 12 (2026), eaed5603, DOI 10.1126/sciadv.aed5603; supplementary Table S1 and source figures are summarized in docs/01_reference_evidence.md and docs/13_candidate_frames_r06.md. ROBOTIS exact product manuals and drawing retrieval outcomes are logged in reports/sources/supplier_completion_r07.md.

Appendix A | Complete r07 BOM index

All 105 unique part numbers and aggregate quantities below are generated directly from bom/bom_r07.csv. Line-item IDs/revisions/roles/mass/provenance in the CSV remain authoritative. A provisional geometry label is not supplier approval.

Assembly group	Exact part number	Qty
Actuators	DYNAMIXEL_XC330_M181_T	5
Actuators	DYNAMIXEL_XL330_M077_T	2
Bearings, clips and spacers	A_OUTER_SPACER_L6.6	2
Bearings, clips and spacers	A_OUTER_SPACER_L8.4	2
Bearings, clips and spacers	BRG_2x4x1p2_PROVISIONAL	64
Bearings, clips and spacers	LEG_r02_BRG_2x4x1p2_PROVISIONAL	48
Bearings, clips and spacers	RETAINING_CLIP_1p7x3p6_T0p6_PROVISIONAL	83
Bearings, clips and spacers	SPACER_2mm_L0.6	18
Bearings, clips and spacers	SPACER_2mm_L1.8	2
Bearings, clips and spacers	SPACER_2mm_L3.0	4
Bearings, clips and spacers	SPACER_2mm_L3.6	2
Bearings, clips and spacers	SPACER_2mm_L9.0	4
Controller board and carrier	OPENRB150_PCB_ENVELOPE_r07	1
Controller board and carrier	OpenRB150_Carrier_r07_EdgeRetainerLeft	1
Controller board and carrier	OpenRB150_Carrier_r07_EdgeRetainerRight	1
Controller board and carrier	OpenRB150_Carrier_r07_M1p4RetainerBolt	4
Controller board and carrier	OpenRB150_Carrier_r07_M1p4SplitClampBolt	4
Controller board and carrier	OpenRB150_Carrier_r07_SplitClampCaps	2
Controller board and carrier	OpenRB150_Carrier_r07_Tray	1
Frames and fixed ties	FIXED_TIE_M2_CORE_L2p8	8
Frames and fixed ties	FIXED_TIE_OD2p4_TAPMINOR1p7_L53p6	4
Frames and fixed ties	FRM_LUMBAR_ROUNDED_CRADLE_L_r07_final	1
Frames and fixed ties	FRM_LUMBAR_ROUNDED_CRADLE_R_r07_final	1
Frames and fixed ties	MOTOR_CENTER_INSULATING_SHIM_T0p8	1
Frames and fixed ties	TORSO_FORE_TRANSVERSE_FRAME_r06	1
Frames and fixed ties	TORSO_HIND_TRANSVERSE_FRAME_r06	1
Legs, pivots and output hardware	LEG_r02_CURVED_CONTACT_PAD	4
Legs, pivots and output hardware	LEG_r02_HEADED_PIN_D1p9_L4p8	4
Legs, pivots and output hardware	LEG_r02_JOURNAL_D1p9_L10p75	4
Legs, pivots and output hardware	LEG_r02_JOURNAL_D1p9_L11p7	4
Legs, pivots and output hardware	LEG_r02_JOURNAL_D1p9_L4p7	8
Legs, pivots and output hardware	LEG_r02_JOURNAL_D1p9_L9p9	2
Legs, pivots and output hardware	LEG_r02_SPACER_D3_ID2p05_L0p6	16
Legs, pivots and output hardware	LEG_r02_SPACER_D3_ID2p05_L2p15	4
Legs, pivots and output hardware	LEG_r02_SPACER_D3_ID2p05_L2p2	4

Assembly group	Exact part number	Qty
Legs, pivots and output hardware	LEG_r02_SPACER_D3_ID2p05_L4p0	4
Legs, pivots and output hardware	LEG_r02_SPACER_D3_ID2p05_L4p2	4
Legs, pivots and output hardware	LEG_r02_SPACER_D3_ID2p05_L5p8	4
Legs, pivots and output hardware	LEG_r02_SPACER_D3_ID2p05_L7p35	4
Legs, pivots and output hardware	LEG_r03_DIRECT_FORE_RETURN_RAIL	2
Legs, pivots and output hardware	LEG_r03_DIRECT_FORE_SHANK	2
Legs, pivots and output hardware	LEG_r03_DIRECT_FORE_THIGH	2
Legs, pivots and output hardware	LEG_r03_DIRECT_HIND_RETURN_RAIL	2
Legs, pivots and output hardware	LEG_r03_DIRECT_HIND_SHANK	2
Legs, pivots and output hardware	LEG_r03_DIRECT_HIND_THIGH	2
Legs, pivots and output hardware	LEG_r03_DUAL_KEYED_H_JOURNAL_D1p9_L10p85	4
Legs, pivots and output hardware	LEG_r03_FORE_BENT_INPUT_HORN_KEYED_SAME_DIRECTION_PROVISIONAL	2
Legs, pivots and output hardware	LEG_r03_FORE_CURVED_FOOT_RETURN_TOWARD_HIP	2
Legs, pivots and output hardware	LEG_r03_HIND_BENT_INPUT_HORN_KEYED_SAME_DIRECTION_PROVISIONAL	2
Legs, pivots and output hardware	LEG_r03_HIND_CURVED_FOOT_RETURN_TOWARD_HIP	2
Legs, pivots and output hardware	LEG_r05_ANKLE_SPACER_OD3_ID2p05_L2p6	2
Legs, pivots and output hardware	LEG_r07_ANKLE_CLAMP_L_OD6_ID2p05_T2p4	1
Legs, pivots and output hardware	LEG_r07_ANKLE_CLAMP_R_OD6_ID2p05_T2p4	1
Legs, pivots and output hardware	LEG_r07_ANKLE_JOURNAL_L_D1p9_L15p05	1
Legs, pivots and output hardware	LEG_r07_ANKLE_JOURNAL_R_D1p9_L15p05	1
Legs, pivots and output hardware	LEG_r07_M1p4_ANKLE_GRUB_CORE1p1_L1p65	2
Legs, pivots and output hardware	TORSO_OUTPUT_JOURNAL_D1p9_L25p8_GROOVED	8
Legs, pivots and output hardware	TORSO_OUTPUT_SPACER_L11.0	4
Legs, pivots and output hardware	TORSO_OUTPUT_SPACER_L19.4	4
Spine drive and joints	DIRECT_CRANK_FLUSH_M2_L	1
Spine drive and joints	DIRECT_CRANK_FLUSH_M2_R	1
Spine drive and joints	GUIDE_COPPER_r02	8
Spine drive and joints	G_CRANK_SPACER_L0p3	2
Spine drive and joints	HORN_GAP_WASHER_OD8_T0p3	2

Assembly group	Exact part number	Qty
Spine drive and joints	HORN_M2_TRIMMED_CORE_L1p9	4
Spine drive and joints	HORN_PATTERN_ASSUMED_OD8_T1p2	2
Spine drive and joints	JOURNAL_D1p9_L8_GROOVED	2
Spine drive and joints	SPINE_r04_A1_STEPPED_CROSS_SHAFT_L85p6	1
Spine drive and joints	SPINE_r04_A2_STEPPED_CROSS_SHAFT_L85p6	1
Spine drive and joints	SPINE_r04_B1_STEPPED_CROSS_SHAFT_L72	1
Spine drive and joints	SPINE_r04_B2_STEPPED_CROSS_SHAFT_L72	1
Spine drive and joints	SPINE_r04_C1_STEPPED_CROSS_SHAFT_L85p6	1
Spine drive and joints	SPINE_r04_C2_STEPPED_CROSS_SHAFT_L85p6	1
Spine drive and joints	SPINE_r04_CAPTURE_CAP_NEG	4
Spine drive and joints	SPINE_r04_CAPTURE_CAP_POS	4
Spine drive and joints	SPN_r02_L2	4
Spine drive and joints	SPN_r02_L3	4
Spine drive and joints	SPN_r02_L4	4
Spine drive and joints	SPN_r02_L5	4
Springs and hind tendon drive	KNEE_r05_M2_CORE_D1p65_L8p8	2
Springs and hind tendon drive	KNEE_r05_M2_HEX_NUT_AF4_T2p8	2
Springs and hind tendon drive	KNEE_r05_REMOVABLE_REAR_LOCK	1
Springs and hind tendon drive	LEG_r02_COMPRESSION_PIVOT_SEAT_NATIVE	8
Springs and hind tendon drive	LEG_r02_COMP_GUIDE_ROD_D1p2_L14	4
Springs and hind tendon drive	LEG_r02_COMP_GUIDE_TUBE_OD1p7_ID1p3_L14	4
Springs and hind tendon drive	LEG_r02_foreL_COMPRESSION_WIRE_L24p73863375	1
Springs and hind tendon drive	LEG_r02_foreL_EXTENSION_WIRE_L17p5917383	1
Springs and hind tendon drive	LEG_r02_foreR_COMPRESSION_WIRE_L24p73863375	1
Springs and hind tendon drive	LEG_r02_foreR_EXTENSION_WIRE_L17p5917383	1
Springs and hind tendon drive	LEG_r02_hindL_COMPRESSION_WIRE_L29p55480766	1
Springs and hind tendon drive	LEG_r02_hindL_EXTENSION_WIRE_L25p61137956	1
Springs and hind tendon drive	LEG_r02_hindR_COMPRESSION_WIRE_L29p55480766	1
Springs and hind tendon drive	LEG_r02_hindR_EXTENSION_WIRE_L25p61137956	1
Springs and hind tendon drive	TENDON_r03_GUIDE_WASHER_OD3p6_ID2p05_Tp2	4
Springs and hind tendon drive	TENDON_r04_DRIVE_OUTER_WASHER_OD3p6_ID2p05_Tp2	1
Springs and hind tendon drive	TENDON_r04_DRIVE_ROOT_SHIM_OD3_ID2p05_Tp2	1
Springs and hind tendon drive	TENDON_r04_D_DRIVE_Rp95_Fp8_L3p7	1
Springs and hind tendon drive	TENDON_r04_GUIDE_OD5p4_ID2p05_L1p2	4
Springs and hind tendon drive	TENDON_r04_HEADED_GUIDE_SHAFT_D1p9_L3p6	4
Springs and hind tendon drive	TENDON_r07_L_CABLE	1
Springs and hind tendon drive	TENDON_r07_M1_ANCHOR_BOLT_COREp75_L2p2_HEAD1p6	2
Springs and hind tendon drive	TENDON_r07_M1p4_TERMINAL_GRUB_CORE1p1_Lp95	2
Springs and hind tendon drive	TENDON_r07_R_CABLE	1
Springs and hind tendon drive	TENDON_r07_STEEL_KEYED_CLAMP_R3p3_6p4_T1p4	2
Springs and hind tendon drive	TENDON_r07_WHEEL_OD13_KEYED_POSITIVE_TERMINALS	1

Appendix totals: 105 unique part numbers, 486 assembly instances. Controller PCB mass remains unknown; the separate 10 g controller + 8 g harness sensitivity case is not booked as a purchase mass.